

Mostafa Jafarian Abyaneh

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SUMMARY

Computational mechanics engineer and PhD student with eight years of nonlinear finite element modeling across composites, concrete, rock, and soil–structure systems. Builds Python-automated FE pipelines and calibrates constitutive models against experimental data. Three peer-reviewed papers in ASCE and ACI journals.

EDUCATION

University of Maryland, College Park

Ph.D. in Civil Engineering

College Park, MD

2026 – Present

Dalhousie University

M.A.Sc. in Civil Engineering | GPA: 3.4/4.0

Halifax, NS, Canada

2017 – 2020

Sharif University of Technology

M.Sc. in Civil / Structural Engineering

Tehran, Iran

2014 – 2017

TECHNICAL SKILLS

- **FEA & Simulation:** Abaqus, Ansys, PLAXIS 3D, LAMMPS, SAP2000, ETABS, AutoCAD
- **Programming:** Python (NumPy, SciPy, Pandas), MATLAB, SQL, Git, LaTeX
- **Methods:** Nonlinear FEA, Constitutive Modeling, Damage Initiation & Evolution, RVE / Micromechanics, Disturbed State Concept, Soil–Structure Interaction, Bayesian Optimization, Uncertainty Quantification, Materials Testing

EXPERIENCE

Doctoral Research Assistant

University of Connecticut

Sep. 2024 – Dec. 2025

Storrs, CT

- Simulated the **elastic–plastic response of randomized CFRP composite RVEs** in Abaqus, resolving damage initiation and evolution at the constituent level.
- Automated the full FE workflow in **Python** — geometry, meshing, stochastic parameter randomization — so many layups and property variations run as one batch.
- Applied **Bayesian optimization** to the material randomization scheme, improving how well the sampled realizations span the property space.
- Stood up **molecular dynamics** capability in LAMMPS for multiscale modeling.

Graduate Research Assistant

Dalhousie University

Sep. 2017 – May 2020

Halifax, NS, Canada

- Built a **nonlinear FE model in MATLAB** for the lateral behavior of concrete-filled FRP tube piles, validated against **full-scale field tests** from a highway bridge on Route 40, Virginia; published in the *ASCE International Journal of Geomechanics*.
- Modeled **axial pile behavior in multi-layered soil in PLAXIS 3D**, including the soil–FRP interface.
- Ran uniaxial compression testing of **tire-derived aggregate concrete** specimens for bridge applications.

Graduate Research Assistant

Sharif University of Technology

Sep. 2014 – Jan. 2017

Tehran, Iran

- Developed a **disturbed state concept constitutive model** for the softening and volumetric response of rock under triaxial compression; published in *ASCE IJG*.
- Ran triaxial and uniaxial test programs across four material types and fitted constitutive parameters to the measured stress–strain curves.

SELECTED PUBLICATIONS

1. **Jafarian Abyaneh, M.**, El Naggar, H., and Sadeghian, P. (2020) “Numerical Modeling of the Lateral Behavior of Concrete-Filled FRP Tube Piles in Sand.” *ASCE Int. J. of Geomechanics*, Vol. 20, No. 8. [doi:10.1061/\(ASCE\)GM.1943-5622.0001725](https://doi.org/10.1061/(ASCE)GM.1943-5622.0001725)
2. **Jafarian Abyaneh, M.** and Toufigh, V. (2018) “Softening Behavior and Volumetric Deformation of Rocks.” *ASCE Int. J. of Geomechanics*, Vol. 18, No. 8. [doi:10.1061/\(ASCE\)GM.1943-5622.0001200](https://doi.org/10.1061/(ASCE)GM.1943-5622.0001200)